Com 307
Machine Learning and Data Mining
Project 2
Naive Bayes classifier for wrongful conviction

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This is a Naive Bayes classifier that estimates P(exonerated | offense, state, race). There are two classes:

- **Exonerated:** cases from the National Registry of Exonerations spreadsheet.
- **Convicted, not exonerated:** federal sentences from the USSC FY2018 datafile.

For each class I counted how often each offense, state, and race shows up, turned the counts into conditional probabilities with Laplace smoothing, and combined them with the class priors. I did the multiplication in log space so tiny numbers wouldn't underflow.

The code is main.py, plus extract_ussc.py for pulling the needed columns out of the USSC file. The probability tables (priors, the three likelihood tables, and the posterior for every offense x state x race combination) are in the three output folders, one per run.

Getting the data into shape

The USSC file has 27,693 columns, but almost all of them are the same few variables repeated. I only needed OFFGUIDE, DISTRICT, MONRACE, HISPORIG, NEWRACE, and MONSEX.

The copy I started with only had 11,464 rows. When I checked the sentencing month, it only covered October to mid-December 2017, so it had been cut off. I downloaded the full FY2018 file from the USSC website instead, which has all 69,425 cases. That file was too big to load into pandas on my laptop (the process got killed for running out of memory), so extract_ussc.py streams it one line at a time and saves just the columns I need to a small CSV.

I dropped 805 of the 69,425 rows (1.2%) for missing or unknown codes, which left 68,620. All 94 districts are present.

Decisions I had to make

1. Decades of exonerations vs. one year of convictions.

I kept exonerations whose conviction year falls between 1989 and 2018 (1,938 of 2,515 cases). The window ends in 2018 because that's the year of my conviction data. The 30-year length was a judgment call - long enough to keep most of the exonerations, but recent enough that 2018 federal convictions are at least somewhat comparable. It drops 577 cases with convictions going back to 1956.

I then treated 2018 as a typical year and multiplied the conviction count by 30 to cover the same window. That assumption is not truly accurate in an obvious way, because federal caseloads and offense mix have changed a lot since 1989. But it only affects the prior, not the likelihood tables.

Another big thing to consider is the fact that exonerations usually come many years after conviction, so people convicted recently haven't had time to be exonerated yet. That makes recent convictions look safer than they are.

2. State vs. federal. This was the biggest issue.

The USSC data is federal only, but about 95% of the exonerations are state cases. So the comparison isn't really "wrongly convicted vs correctly convicted." It's closer to "state cases that got exonerated vs federal cases."

To deal with this I ran the model two ways:

- with all exonerations (1,932 after cleaning), and
- with only the federal exonerations (107 after cleaning), which is a fair comparison but a very small sample.

3. Matching offenses

The registry uses descriptions like "Murder" or "Drug Possession or Sale," and the USSC uses 30 OFFGUIDE codes. I built a crosswalk by hand. The calls I wasn't sure about:

- **Drug Possession or Sale:** I sent this to Drug Trafficking (10) rather than Drug Possession (9), since most federal drug cases are trafficking.
- **Attempted Murder:** sent to Assault, because the federal guideline for it (2A2.1) is in the assault section, and OFFGUIDE is based on the guideline.
- **Child Abuse, Menacing, Other Violent Felony:** sent to Assault.
- **Sex Offender Registration:** sent to Sex Abuse, because the federal guideline for failure to register (2A3.5) sits in the sexual-abuse part of the guidelines.
- **Accessory to Murder, Perjury, False Report:** sent to Administration of Justice.
- **Solicitation, Traffic, Destruction of Property, and the various "Other" labels:** sent to Other.

4. Matching locations

The USSC records 94 federal districts, while the registry records state and county. I turned districts to states, so all four Texas districts became "Texas," and mapped the registry's "Fed-XX" labels to the same state names. I dropped the 6 military exonerations because they don't have a state.

5. Matching race

The assignment says to use MONRACE, but MONRACE doesn't have a Hispanic category; Hispanic ethnicity is stored separately in HISPORIG. The registry treats Hispanic as a race. Using MONRACE would have counted most Hispanic defendants as White. So I used NEWRACE, which combines the two into White, Black, Hispanic, and Other. I converted the registry's Asian and Native American categories into Other to match.

6. Convictions minus exonerations

I checked for federal exonerations of people convicted in 2018 so I could subtract them. There were none, so nothing changed.

Findings

	All exonerations	**Federal only (α = 1)**	**Federal only (α = 0.1)**
Exonerations used	1,932	107	107
Prior P(exonerated)	0.00094	0.000052	0.000052
Murder likelihood ratio	72.8	7.9	8.0
Black likelihood ratio	2.29	0.96	0.95
White likelihood ratio	1.75	2.60	2.64
Hispanic likelihood ratio	0.24	0.28	0.28

P(exon | Murder, Illinois, Black)	0.52 (557x base)	0.00093 (18x base)	0.0012 (23x base)
P(exon | Fraud, New York, White)	0.0014 (1.5x base)	0.00092 (18x base)	0.0016 (30x base)

With all exonerations

The model says murder, Illinois, and being Black are the strongest predictors of exoneration. A Black defendant convicted of murder in Illinois gets P(exonerated) = 0.52, which is 557 times the base rate.

I don't think that means much. Murder is mostly a state crime, so there were only 318 federal murder cases in all of 2018, against 663 murder exonerations.

With only federal exonerations

Most of that goes away.

- The murder ratio drops from 73 to about 8.
- The Black effect disappears completely (ratio of about 1).
- White defendants become overrepresented instead.

The clearest sign of the shift: in the all-exonerations run, Fraud/New York/White is barely above the base rate (1.5x), while Murder/Illinois/Black is at 557×. In the federal run with α = 0.1, they flip. Fraud/New York/White is at 30x and Murder/Illinois/Black is at 23x. Which profile looks "most likely to be exonerated" depends almost entirely on which exonerations you compare against.

Hispanic defendants are underrepresented in both runs

(about 0.24-0.28). I think this is mostly about immigration.

Smoothing

My first federal-only run used α = 1, and some of the top-ranked categories had zero exonerations.

- **Offenses -** Burglary (12.2x), Food and Drug (11.4x), Antitrust (9.3x), Individual Rights (7.4x)

- **States and territories -** Northern Mariana Islands (28.3x), Guam (9.4x), Virgin Islands (7.4x), Delaware (6.1x)

That's a smoothing artifact. With only 107 exonerations across about 30 offense types, adding 1 to a zero count makes it look like 0.7% of exonerations. That's much more than burglary's 0.06% share of federal convictions, so a category with no exonerations at all ends up looking 12 times more likely.

Use of generative AI

I used Claude for assistance in this project:

- It helped me understand the assignment and the USSC codebook.
- It wrote the district-to-state and crime-to-OFFGUIDE lookup tables.
- Fixing some bugs in my code.
- Especially helped me fix the posterior function using numpy